

Xuewen (Daphne) Yang

✉ xuewenyang1@gmail.com | ☎ (415) 298-2845 | 📍 San Francisco Bay Area, CA
<https://www.linkedin.com/in/xuewen-daphne-yang/> | <https://www.xuewenyang.xyz>

Software Engineer/Data Engineer with 2 YOE building production-grade ETL pipelines, distributed data processing systems, and automated validation frameworks at scale. Experienced in Python, SQL, Scala, Spark, and Airflow, with a focus on data integrity, observability, and enabling data-driven decision-making across complex engineering workflows.

SKILLS

Python, SQL, Scala, Apache Spark, Hive, Iceberg, Pandas, BigQuery, Tableau, Looker Studio, Data Analytics, Data Mining, Data Visualization, Machine Learning, Google Cloud Platform (GCP), Google Cloud Storage (GCS), Apache Airflow, Linux, Git, Jupyter Notebook, Jira, Java, C/C++

WORK EXPERIENCE

Software Engineer/Data Engineer, Walmart Global Technology, Sunnyvale, CA July 2024 - Present

- Developed and maintained **Scala/Spark ETL pipelines orchestrated by Airflow**, processing **10M+ daily events** from **Hive** and **Iceberg tables/views**, transforming transactional and user activity data for sponsored product attribution and reporting.
- Built new data pipelines to ingest and join transactional, user activity, and page metadata datasets, expanding analytics coverage across the advertising platform.
- Developed an automated **Scala-based data validation framework**, reducing validation time from **4+ hours to under 10 minutes** while detecting schema drift and data quality issues before production.
- Developed attribution metrics and built Looker dashboards to monitor data quality and reporting health.
- Maintained **10+ Airflow-orchestrated production workflows**, performing on-call incident response and root cause analysis to ensure pipeline reliability.

Software Engineer Intern, Walmart Global Technology, Sunnyvale, CA June 2023 - Aug. 2023

- Built **3 Tableau dashboards** comprising **15+ visualizations** for advertising analytics and campaign performance monitoring, enabling self-serve reporting for cross-functional stakeholders.
- Analyzed **10B+ row attribution datasets** using SQL and Python to surface business insights and support advertising reporting needs.

Advancement Data and Business Intelligence Assistant, UCSD, La Jolla, CA Nov 2022 - Dec 2023

- Maintained constituent datasets and supported data management operations for university fundraising initiatives.
- Generated reports, managed exception handling workflows, and supported operational data validation tasks.
- Assisted with reporting automation and data maintenance processes.

EDUCATION

Master in Data Science (Part-time) La Jolla, CA
University of California San Diego 09/2024 - 03/2026

GPA: 3.9/4.0 MDS Supplemental Scholarship

B.S. in Mathematics - Computer Science & B.S. in Data Science La Jolla, CA
University of California San Diego 09/2020 - 03/2024

GPA: 3.89/4.0 Mae Brown Scholarship, Errett Bishop Scholarship, Provost's Honor, Cum Laude

PROJECTS

Image-Based Mushroom Edibility Classification with Deep Learning - MDS Capstone Project

- Built an end-to-end image classification pipeline on a 100K+ image dataset, including data ingestion, preprocessing, duplicate removal, and class imbalance handling.
- Developed reproducible training and evaluation workflows to benchmark ResNet-18, ResNet-50, and Swin Tiny Transformer architectures.
- Improved model reliability through decision threshold optimization, achieving **92.26% accuracy** and **94.02% toxic recall** while reducing high-risk false negatives.
- Conducted large-scale model evaluation and comparative analysis, identifying a fine-tuned Swin Tiny model as the optimal solution for safety-critical classification tasks.

Scoring Risk of Default Using Banking Transaction Data - Undergrad DS Capstone Project

- Developed a machine learning-based credit risk scoring model for first-time loan and credit card applicants using banking transaction data.
- Achieved **84% accuracy** and **0.87 AUC** with an XGBoost model, effectively ranking borrower risk for credit assessment.
- Analyzed feature importance to identify key default risk factors and support lending decisions.